

Conditional and Joint Probability

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2025-06-19

Exercise 1

A small survey was conducted with 50 people regarding whether they own a car and whether they have a driver's license.

Data (Example)

You can modify the values below or use the following as an example to work with:

```
# Example frequency table:
datos <- data.frame(
  car = rep(c("Si", "Si", "No", "No"), 1),
  licence = rep(c("Si", "No", "Si", "No"), 1),
  frequency = c(20, 15, 5, 10)
)
```

```
# Total people
total <- sum(datos$frequency)
cat("Total people: ", total)
```

```
## Total people: 50
```

1. Joint probability that a person has a car and a driver's license.

```
#Sum of cases (for later)
freq_both <- datos$frequency[datos$car == "Si" & datos$licence == "Si"]
joint_prob <- freq_both / total
```

```
#Probability
cat("Probability that a person has both car and license: ",
    joint_prob*100, "%")
```

```
## Probability that a person has both car and license: 40 %
```

2. Conditional probability that a person has a driver's license given they have a car.

```
#Sum of cases (for later)
freq_car <- sum(datos$frequency[datos$car == "Si"])
freq_licence <- sum(datos$frequency[datos$licence == "Si"])
```

```
#Probability
prob_cartolicense <- (freq_both/freq_car)
cat("Total people who has a licence given they have a car: ",
    prob_cartolicense*100, "%")
```

```
## Total people who has a licence given they have a car: 57.14286 %
```

3. Conditional probability that a person has a car given they have a driver's license.

```
#Probability
```

```
prob_licensetocar <- (freq_both/freq_license)
cat("Total people who has a licence given they have a car: ",
prob_licensetocar*100, "%")
```

```
## Total people who has a licence given they have a car: 80 %
```

4. Check whether owning a car and having a driver's license are independent events.

```
#Probability
```

```
prob_car=freq_car/total
prob_licence=freq_license/total
```

```
#If because why not
```

```
if (prob_car == prob_licensetocar & prob_licence == prob_cartolicense) {
  cat("The events are independent")
} else {
  cat("The events are dependent")
}
```

```
## The events are dependent
```

5. Create a Venn diagram representing the sets from the exercise and their intersection

```
library(VennDiagram)
```

```
## Cargando paquete requerido: grid
```

```
## Cargando paquete requerido: futile.logger
```

```
venn.plot <- draw.pairwise.venn(
  area1 = freq_car,
  area2 = freq_license,
  cross.area = freq_both,
  category = c("Car", "License"),
  fill = c("#953472", "#126593" ),
  lty = "black",
  cex = 2,
  cat.cex = 2,
  cat.col = c("#311225", "#063045")
)
```

